

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	55.0 / Female
Department	General Surgery
Diagnosis	Bladder infection
UTI Classification	Complicated UTI
Sample Type	Suprapubic Aspirate

Clinical Presentation

Chief Complaints: Lower abdominal pain;Dysuria

Comorbidities: Diabetes

Surgical History: Hysterectomy

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	31.0	%	20 - 40
Wbc	9600.0	cells/ μ L	4000 - 11000
Polymorphs	63.0	%	40 - 75
Crp	12.0	mg/L	< 10
Rft Serum Creatinine	1.1	mg/dL	0.6 - 1.3
Serum Uric Acid	4.5	mg/dL	3.5 - 7.2
Blood Urea	27.0	mg/dL	7 - 20
Cue Pus Cells	12.0	/HPF	0 - 5
Epithelial Cells	4.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	2.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Candida glabrata
Bacteria Type	Fungus (Yeast; often shows high resistance to fluconazole)

Antibiotic Susceptibility Profile

Predicted Sensitive	Amphotericin B, Voriconazole
Predicted Resistant	Fluconazole

Recommended Therapy

Amphotericin B (deoxycholate)

Dosage: 0.5 mg/kg IV infusion over 1-2 hours daily (e.g., 25 mg for a 55-kg patient).

Precautions: Monitor serum creatinine and electrolytes daily; prehydrate with 0.9% saline 1-2 hours before infusion; consider liposomal formulation if renal function worsens. Avoid concurrent nephrotoxic drugs. Monitor blood glucose due to diabetes.

Rationale: Candida glabrata is often resistant to fluconazole; amphotericin B provides broad fungicidal activity against this organism. It is effective for invasive urinary tract infections and is a first-line agent when other azoles are contraindicated.

Voriconazole

Dosage: Oral 200 mg twice daily for 24 hours, then 100 mg twice daily; or IV 6 mg/kg twice daily if oral not tolerated.

Precautions: Check liver function tests before and during therapy; monitor for visual disturbances and photosensitivity. Avoid concomitant strong CYP3A4 inhibitors/inducers. Adjust dose if hepatic impairment or if patient is on medications metabolized by CYP3A4. Monitor blood glucose due to diabetes.

Rationale: Voriconazole is active against Candida glabrata and offers a convenient oral regimen. It is a suitable alternative when amphotericin B is contraindicated or not tolerated, providing effective systemic coverage for complicated urinary tract infections.

Antibiotic Drug History

Clinical Summary

Clinical Summary

- **Infection:** Complicated lower urinary tract infection (bladder) in a 55-year-old diabetic female.
- **Pathogen:** *Candida glabrata* (yeast).
- **Resistance profile:** Resistant to fluconazole.
- **Sensitivity profile:** Susceptible to amphotericin B and voriconazole.

Recommended Therapy

1. Amphotericin B (deoxycholate)

- **Dose:** 0.5 mg kg⁻¹ IV over 1-2 h daily (≈25 mg for a 55-kg patient).
- **Rationale:** Broad-spectrum fungicidal activity; first-line for invasive candidiasis when azoles are ineffective.
- **Precautions:** Daily serum creatinine and electrolytes; pre-hydrate with 0.9 % saline 1-2 h before infusion; avoid concurrent nephrotoxins; monitor and control blood glucose due to diabetes.

2. Voriconazole

- **Dose:** Oral 200 mg BID for 24 h, then 100 mg BID (or IV 6 mg kg⁻¹ BID if oral not tolerated).
- **Rationale:** Potent activity against *C. glabrata* with convenient oral administration.
- **Precautions:** Baseline and periodic liver function tests; watch for visual disturbances, photosensitivity; avoid strong CYP3A4 inhibitors/inducers; adjust dose for hepatic impairment; monitor glucose.

Key Considerations

- No prior antibiotics, so resistance likely driven by organism biology, not prior exposure.
- Diabetes increases risk of nephrotoxicity and hyperglycemia; adjust monitoring accordingly.
- Amphotericin B provides immediate fungicidal coverage; voriconazole offers a less toxic, oral alternative when tolerated.

This regimen targets the identified pathogen while balancing efficacy and safety in a diabetic patient with compromised renal function.